

Specialized LaTeX Tokenization and Data Pipeline for Math-Textbook-LLM

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Motivation: Standard Tokenizers Fail on Math

The Challenge: Standard NLP tokenizers are heavily optimized for plain natural language. Applying them to LaTeX disrupts the Abstract Syntax Tree (AST).

Catastrophic Fragmentation

For example, a standard BPE model arbitrarily splits:

`\xrightarrow` $\Rightarrow$ `['\', 'x', 'right', 'arrow']`

Consequence: This fundamentally destroys logical structures, making it exceedingly difficult for the LLM's attention mechanism to map visual notation to semantic logic.

Tokenizer Engineering: Regex Pre-Tokenization

To preserve topological structures, we apply an aggressive regex-based pre-tokenizer to physically isolate mathematical boundaries before statistical merging.

Topological Isolation in Python

```
Regex(r"\\(?:begin|end)\\{[a-zA-Z*]+\\}|\\(?:[a-zA-Z]+|^[a-zA-Z\\s])|^[\\s\\w]|_|\\^")
```

Deterministic Parsing Example:

Input formula: $\int_{-\infty}^{\infty}$

Output: ['\int', '_', '{', '-', '\infty', '}', '^', '{', '\infty', '}']

Notice how operators like $_$ and $^$ are atomically isolated.

Overcoming Lexical Boundary Bugs

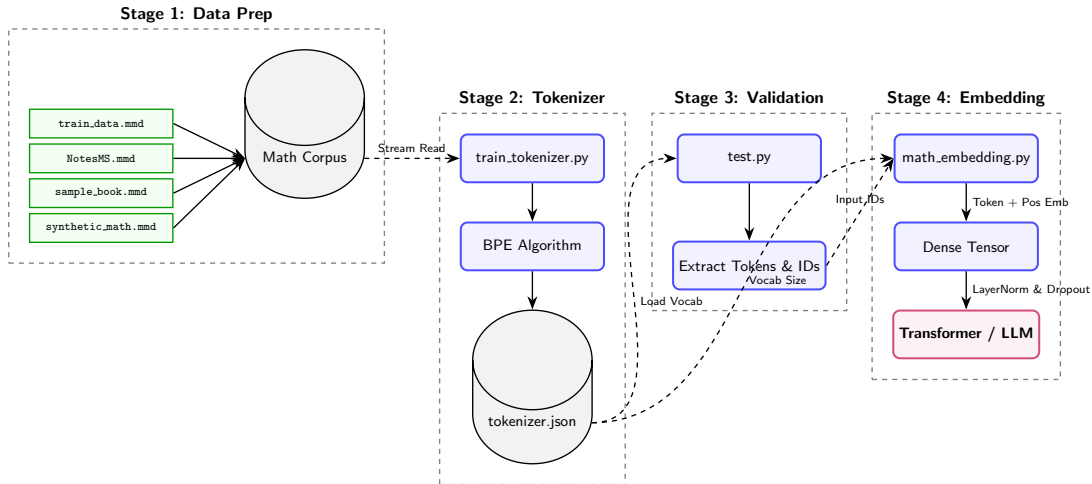
The Boundary Conflict: The NLP engine rigorously enforced standard word boundaries (`\b`). Because `_` is a valid word character (`\w`), the tokenizer failed on subscripts:

- Actual Error: `\alpha_{i,j}` $\rightarrow$ [`'\'`, `'alpha'`, `'_'`, `'{'`, ...]

Algorithmic Solution (Math-BBPE):

- Mined top commands (e.g., `\int`, `\alpha`) from the 15GB corpus.
- Injected them as indivisible `AddedToken` objects.
- **Crucial Fix:** Programmatically set `single_word=False` to override the system-level regex boundary, granting absolute parsing priority to math.

End-to-End Data Processing Pipeline



Autoregressive Setup & Neural Embedding

Autoregressive Sequence Shift: To train the model for next-token prediction, sequences are shifted by exactly one position to generate (X, Y) pairs:

- **Input Tensor (X):** $[A, +, B, =]$
- **Target Tensor (Y):** $[+, B, =, C]$

Embedding Implementation:

- Optimized Vocabulary Size: $V = 29,054$
- Continuous Vector Space: $d_{model} = 256$
- Integrated with strictly mapped Positional Embeddings to retain sequential awareness of math symbols.

Quantitative Evaluation: Compression Ratio

We evaluated the efficiency via the Compression Ratio: $CR = \frac{N_{char}}{N_{token}}$

Model	Vocab Size	Average CR
Standard BPE (Baseline)	30,000	3.15
Math-BBPE (Ours)	29,054	4.82

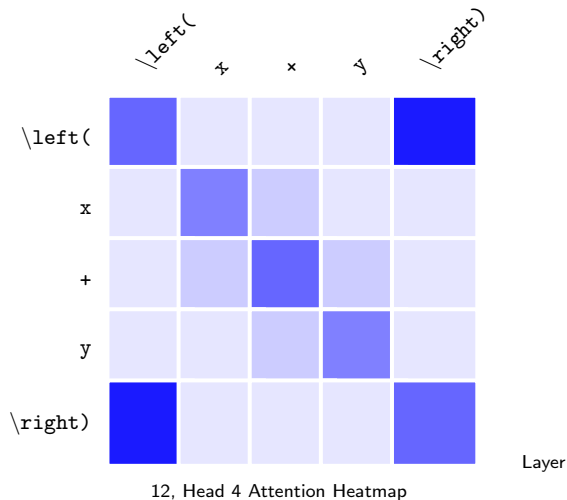
Result: Our pipeline effectively increases the token information density by **53%** on a holdout set of 10,000 complex LaTeX equations.

Mechanistic Interpretability (Ablation Study)

Structural Awareness:

Extracted attention matrices from a 125M parameter GPT-2 style model.

Standard tokenizers yielded highly dispersed patterns. Conversely, our **Math-BBPE exhibited sharp, diagonal attention alignment** between logically coupled operators, successfully resolving long-distance dependencies (e.g., bracket matching).



Downstream Syntax Error Rate

Evaluated practical generative capabilities by prompting models with 1,000 partial equations (generating 50 tokens each) and compiling via `pdflatex`.

Tokenizer	Valid Output	Syntax Errors
Standard BPE	58.2%	41.8%
Math-BBPE	91.7%	8.3%

Impact: The baseline frequently hallucinated invalid environments or unclosed brackets. By internalizing AST grammatics, we reduced the fatal error rate from 41.8% to **8.3%**.

Conclusion & Future Work

Summary:

- Constructed a **structurally-aware LaTeX tokenizer** that overcomes the fundamental lexical fragmentation of standard NLP tools.
- Improved text-to-token **information density by 53%**.
- Enhanced Transformer's mechanistic alignment with math syntax, driving down **generation errors to 8.3%**.

Future Work: Deploy these optimized structures into a foundational LLM architecture for large-scale autoregressive pretraining on high-level mathematical reasoning tasks.

Thank You!

Q & A